

Catching the UWB Pulse

The short duration of UWB waveforms lend them some unique properties. In communications, for example, UWB pulses can be used to provide extremely high data rate performance in multi-user network applications. For radar applications, these same pulses can provide very fine range resolution and precision distance and/or positioning measurement capabilities.

These short duration waveforms are relatively immune to multipath cancellation effects as observed in mobile and in-building environments. Multipath cancellation occurs when a strong reflected wave—for example, off of a wall, ceiling, vehicle, etc—arrives partially or totally out of phase with the direct path signal, causing a reduced amplitude response in the receiver. With very short pulses, the direct path has come

and gone before the reflected path arrives and no cancellation occurs. As a consequence, UWB systems are particularly well suited for high-speed, mobile wireless applications. Also, because of the extremely short duration waveforms, we can implement packet burst and time division multiple access (TDMA) protocols for multi-user communications.

As bandwidth is inversely related to pulse duration, the spectral extent of these waveforms can be made quite large. With proper engineering design, the resultant energy densities (that is, transmitted Watts of power per unit Hertz of bandwidth) can be quite low. This low energy density translates into a low probability of detection (LPD) RF signature. An LPD signature is of particular interest for military applications (example, for

covert communications and radar), however, an LPD signature also produces minimal interference to proximity systems and minimal RF health hazards, significant for both military and commercial applications.

Among the most important advantages of UWB technology, however, are those of low system complexity and low cost. UWB systems can be made nearly 'all-digital', with minimal RF or microwave electronics. Because of the inherent RF simplicity of UWB designs, these systems are highly frequency adaptive, enabling them to be positioned anywhere within the RF spectrum. This feature avoids interference to existing services, while fully utilising the available spectrum.

According to one estimate, 20,000 people could talk on UWB cellphones within one square block with no interference.

year, Intel demonstrated a prototype UWB system that can send data at 100 Mbps and says it plans to push the technology to 200 Mbps and then 400 Mbps, and finally to 500 Mbps. At that speed, it will be 500 times faster than Bluetooth and nine times the speed of next-generation wireless LAN systems.

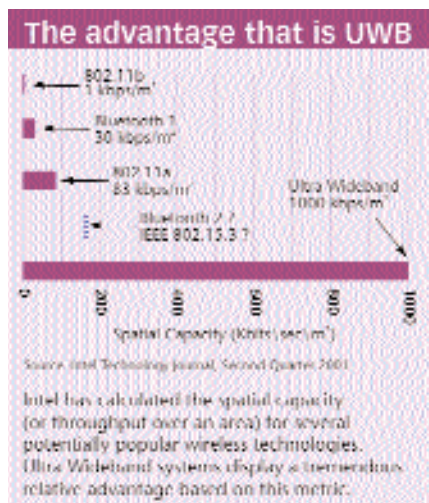
UWB has enough bandwidth to wirelessly network every consumer electronics device in a home from a single access point. This could include the wireless distribution of TV signals (removing the need for cables between set-top boxes around the home), data networking for PCs, PDAs and peripheral devices, home entertainment center components, white appliances, home robotics, home security, baby monitors and more.

Future applications, says Boll, include embedding UWB into battery-powered products such as camcorders, digital cameras, phones, etc so that consumers can wirelessly link images to the UWB-enabled display of their choice, such as a large screen plasma/DLP/LCD, a PC or an actual TV. The ability to embed UWB into a battery-powered device goes back to the low power consumption inherent in ultra wideband.

There will also be UWB-based tracking systems to pinpoint the location of equipment on huge factory floors, find equipment inside hospitals, or locate shipping containers on a dock, among other tasks that are hard for industry to handle now. UWB tracking devices might also be used to guide rescue robots, or even direct lawn mowers.

What will it lead to?

Comparisons with Bluetooth and 802.11 technologies are inevitable. However, some analysts believe that UWB is actually complementary to these technologies. Bluetooth was originally designed as a technology for wireless headsets and was never intended for multimedia applications that require a high data rate.



Watkins believes WiFi is a great technology for data-centric applications, and that it will continue to be very successful. Ultra wideband, by contrast, is ideal for multimedia, due to its high data rate, low cost and low power consumption, and thus, is intended to serve a portion of the market that is today unserved: wireless multimedia connectivity in the home.

Other analysts feel that Bluetooth is a

low-power option for transmitting data over short distances, but is very slow at less than 1 Mbps. All three members of the 802.11 family consume too much power and are too expensive for battery-powered devices. With UWB having the raw speed to swap video and audio files, Bluetooth would be restricted to a handy wireless-form of 'wiring' for low-data-rate connections between computers and their peripherals. And 802.11b, also called WiFi, is becoming popular for wireless computer networks in homes, offices and public spaces.

UWB technology consumes much less power than 802.11, making it ideal for use in battery-powered devices such as cameras and cell phones. WiFi, in contrast, is limited to PCs and things that you can plug into a wall.

UWB's future will include tracking devices for personal items such as keys and eyeglasses; smarter air bags that precisely locate a seat occupant in order to reduce the risk of injury; dual-mode cellphones that improve efficiency and capacity indoors; hospital communications systems that can locate and transmit data between medical devices; and security nets that spread out like an invisible dome over a home or a car, warning of intruders.

UWB products are expected to hit the shelves in the fourth quarter of 2003. This could include systems to transfer audio and video from a stereo or DVD player or camcorder to other devices. Now that will be an entertaining watch. ■

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